| | |
|---|---|
| **Semester** | B.E. Semester VII |
| **Subject** | WAS LAB |
| **Subject Professor In-charge** | Prof. Rakshak Sood |
| **Laboratory** | Online Teams |

| | | |
|---|---|---|
| **Student Name** | Yashraj Patil | |
| **Roll Number** | 23102A0071 | |
| **Grade and Subject Teacher's Signature** | | |

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|---|---|
| **Assignment Number** | 09 – Student Design Experiment |
| **Experiment Title** | Detecting Web Application Reconnaissance via DNS Log Analysis Using Splunk SIEM |
| **Github Link** | https://github.com/Yashraj552/detecting-web-app-recon-splunk |
| **Description** | This lab loads DNS logs into Splunk and uses simple search queries to spot signs of web reconnaissance, such as subdomain scanning, sudden spikes in queries, and lookups of suspicious domains. Suspicious domains are then checked against online threat intelligence tools like VirusTotal. |
| **Step Wise Execution** | **1. Introduction to DNS and DNS Logging**<br><br>The Domain Name System (DNS) is a hierarchical, distributed naming system that translates human-readable domain names (such as www.example.com) into IP addresses that computers use to communicate. It works over port 53, using UDP for most queries and TCP for large responses and zone transfers. Almost every network activity, from browsing a website to a malware callback, starts with a DNS lookup.<br><br>A DNS log is a record of these queries and responses. Each entry typically contains:<br><br>1. Timestamp: when the query occurred.<br>2. Source IP (src_ip): the client that made the request. |

3. Destination IP (dest_ip): the DNS server or resolver that received it.
4. Domain name (FQDN): the Fully Qualified Domain Name being looked up.
5. Query type: the record requested, such as A (IPv4), AAAA (IPv6), MX (mail), TXT, or AXFR (zone transfer).
6. Response code: the outcome, such as NOERROR (success), NXDOMAIN (domain does not exist), or REFUSED (request denied).

Because DNS traffic is rarely blocked and is generated by every host, DNS logs give defenders strong visibility into network behaviour. Attackers also exploit DNS for reconnaissance, command-and-control (C2) communication, data exfiltration and malware delivery.

**2. Web Application Reconnaissance**

Reconnaissance is the first phase of an attack, in which the attacker gathers information about a target before attempting exploitation. For web applications, DNS-based reconnaissance includes:

1. Subdomain enumeration (brute-forcing): the attacker queries many guessed names (admin, dev, staging, api, vpn) under a target domain to discover hidden or forgotten services. Most guesses fail, so this produces a large number of queries from one source with many NXDOMAIN responses.
2. Zone transfer attempts (AXFR): the attacker asks a DNS server to hand over its entire zone file, which would reveal all hostnames. Properly configured servers refuse these requests.
3. Lookups of malicious domains: infected hosts may repeatedly resolve known C2 or phishing domains.

In DNS logs these activities show up as abnormal query volume, a high ratio of failed lookups, many unique subdomains from a single host, and queries to unfamiliar domains.

**3. SIEM and Splunk**

A Security Information and Event Management (SIEM) system collects, normalizes, stores and analyzes log data from many sources

in one place. It supports threat detection, alerting, incident investigation and compliance reporting.

Splunk Enterprise is a widely used SIEM and data analytics platform. Its main functions are:

1. Data ingestion: logs can be uploaded as files, monitored from directories, or received from forwarders.
2. Indexing: incoming data is parsed into individual events, timestamped and stored in an index for fast searching.
3. Search and analysis: data is queried using the Search Processing Language (SPL).
4. Visualization and alerting: results can be shown as tables, charts and dashboards, or trigger alerts.

**4. Key Splunk Concepts Used in the Lab**

1. Index: the repository where Splunk stores ingested data. The wildcard index=* searches all indexes.
2. Source type: a label that tells Splunk how to parse the data (timestamp format, line breaking, field extraction). Here the source type is dns_sample.
3. Event and raw data (_raw): each log line becomes an event, and _raw holds its original text.
4. Fields: named values extracted from events, such as src_ip, fqdn and response_code. Splunk can extract them automatically (for key=value data) or through regular expressions.
5. Time range picker: limits a search to a period. “All time” was used so that all uploaded events are visible regardless of their timestamps.

| Commad | Purpose |
|---|---|
| index=* sourcetype=dns_sample | Retrieves all events of the DNS source type from all indexes. |
| regex | Filters events whose _raw text matches a regular expression. |

| | |
|---|---|
| | The pattern (?i)\b(dns\|domain\|query\|response\|port 53)\b is case-insensitive ((?i)), uses word boundaries (\b), and matches any of the listed DNS-related keywords. |
| stats count by fqdn | Aggregates events and counts how many times each domain was queried. Domains with unusually high counts may indicate scanning, beaconing or abuse. |
| top fqdn, src_ip | Returns the most frequent combinations of domain and source IP, along with count and percentage. It helps identify which hosts generate the most requests and to which domains. |
| timechart span=1h count | Plots event counts over time in fixed intervals, making sudden bursts of automated activity visible. |
| fqdn="maliciousdomain.com" | A field-value search that isolates every event involving a specific suspicious domain, so the affected hosts and activity can be investigated. |

**6. Anomaly Detection in DNS Logs**

Anomaly detection means comparing activity against a normal baseline. Common indicators in DNS logs include:

1. A sudden spike in query volume from one host.
2. A large number of NXDOMAIN (failed) responses.
3. A high number of unique subdomains queried under one parent domain.

4. Zone transfer (AXFR) requests, especially those that are refused.
5. Queries to rarely seen, newly registered or randomly generated domains.
6. Regular, repeating queries at fixed intervals, which may indicate malware beaconing.

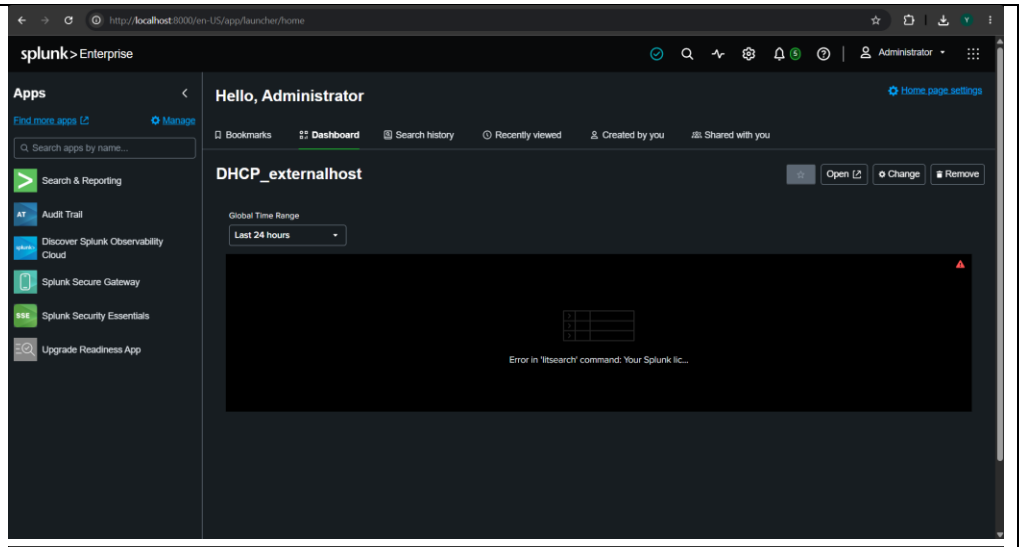
**7. Threat Intelligence and Domain Reputation**

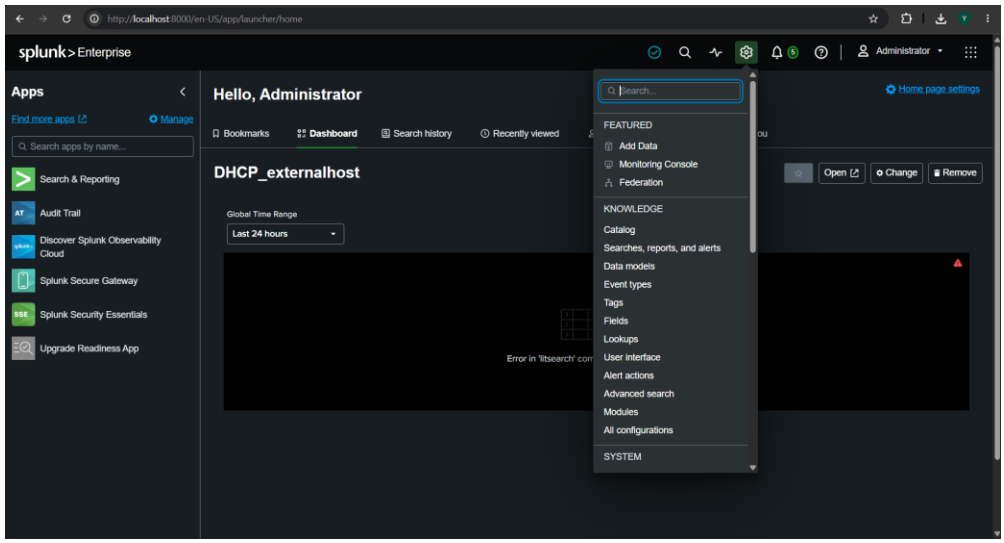
Threat intelligence is information about known malicious indicators, such as domains, IP addresses and file hashes. When an unfamiliar domain appears in the logs, analysts cross-check it against reputation services such as **VirusTotal**, which aggregates verdicts from many security vendors. A domain flagged as malicious, together with the internal hosts that queried it, helps confirm a compromise and decide on containment steps such as blocking the domain or isolating the host.

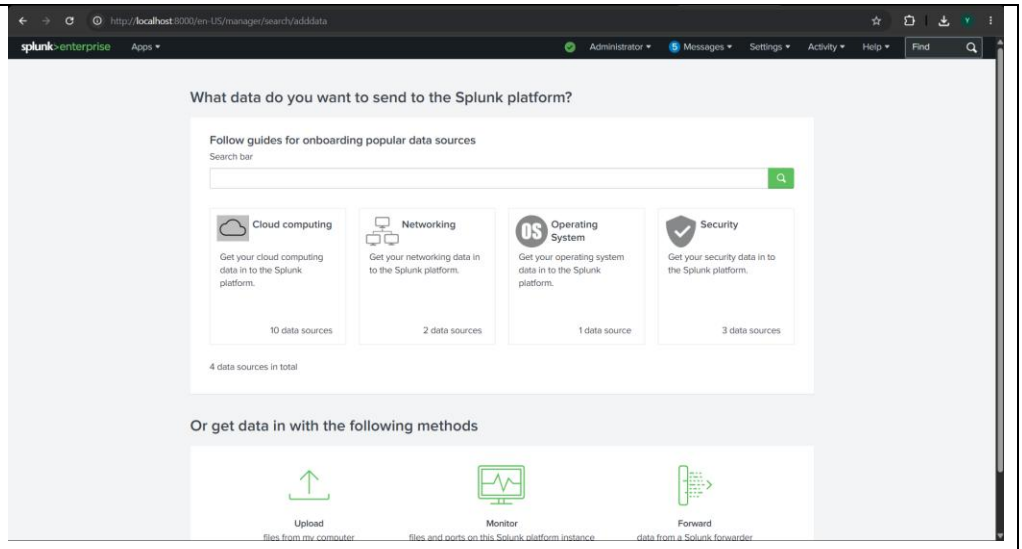
**8. Importance of DNS Monitoring**

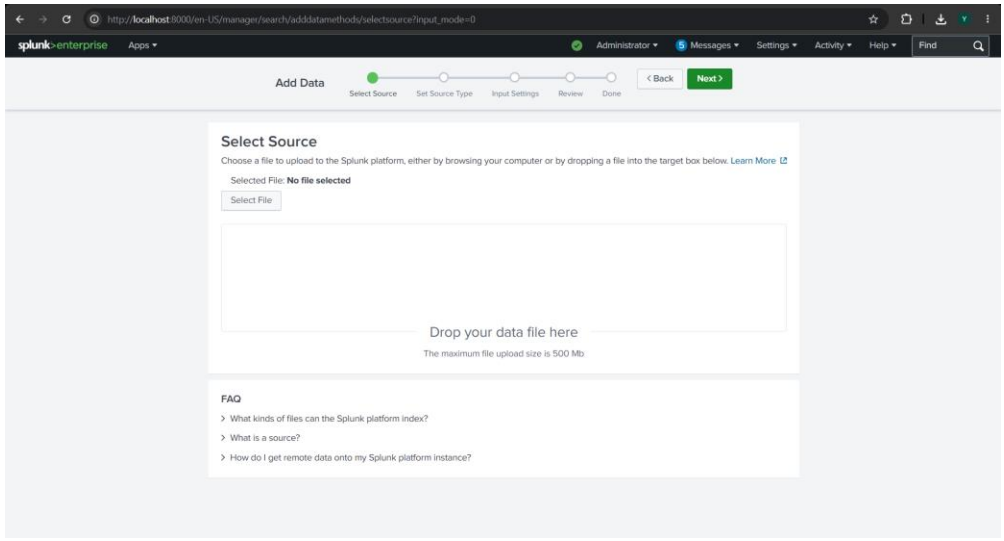
DNS monitoring offers early detection, because reconnaissance and C2 activity often appear in DNS before any other evidence. It also supports forensic investigation, since logs show which hosts contacted which domains and when. Finally, it raises security awareness: a SIEM turns raw, unstructured logs into actionable intelligence for a Security Operations Center (SOC).

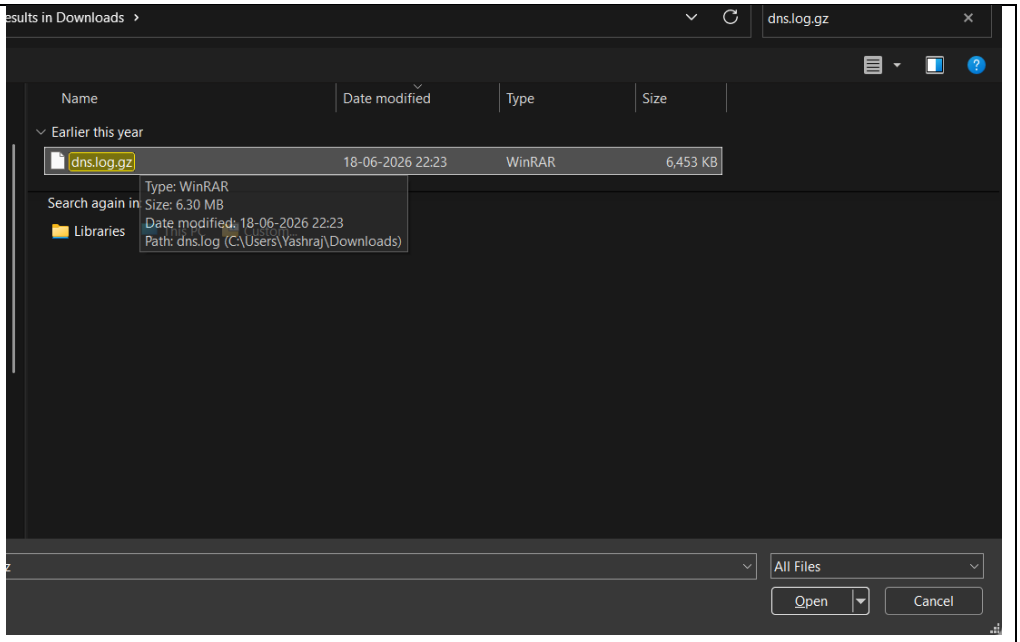
**Screenshots**

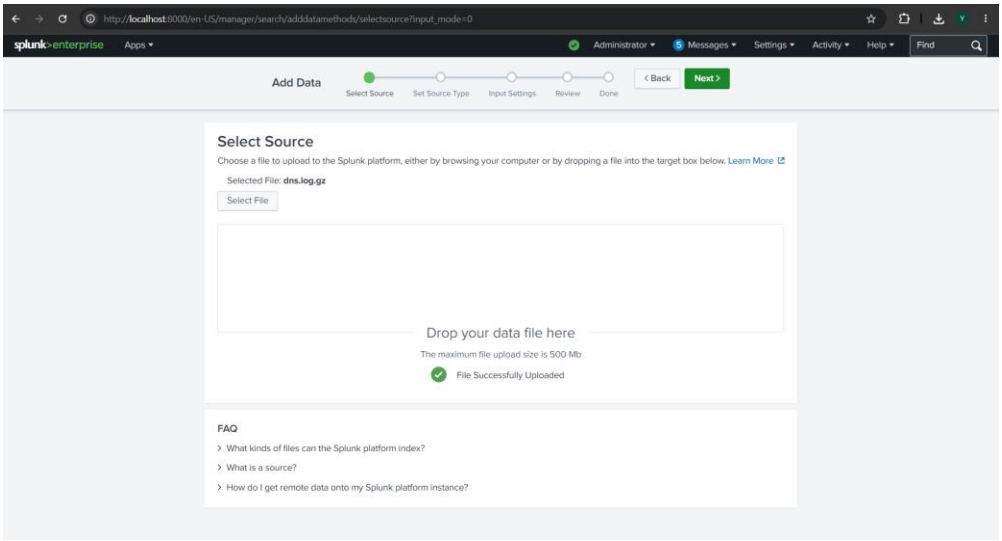


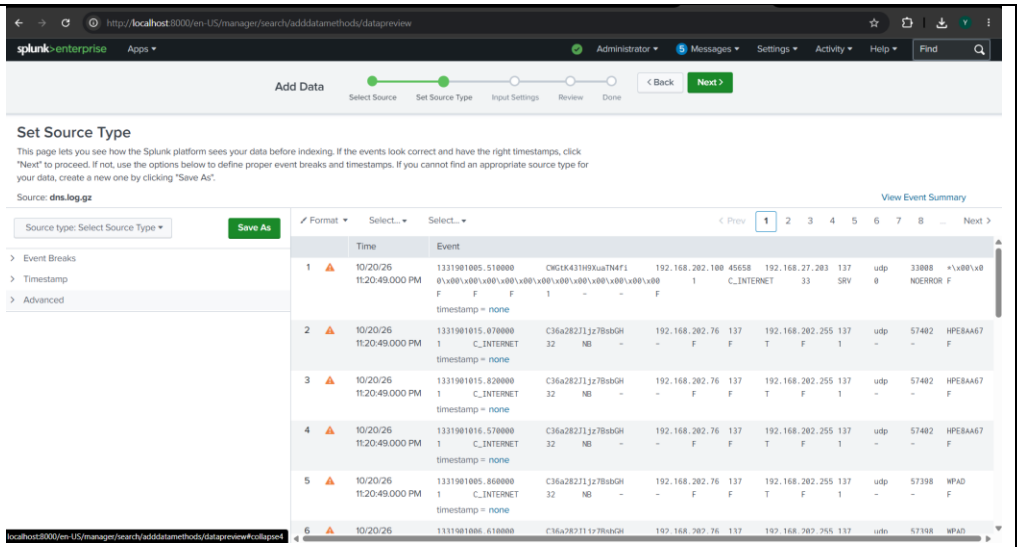



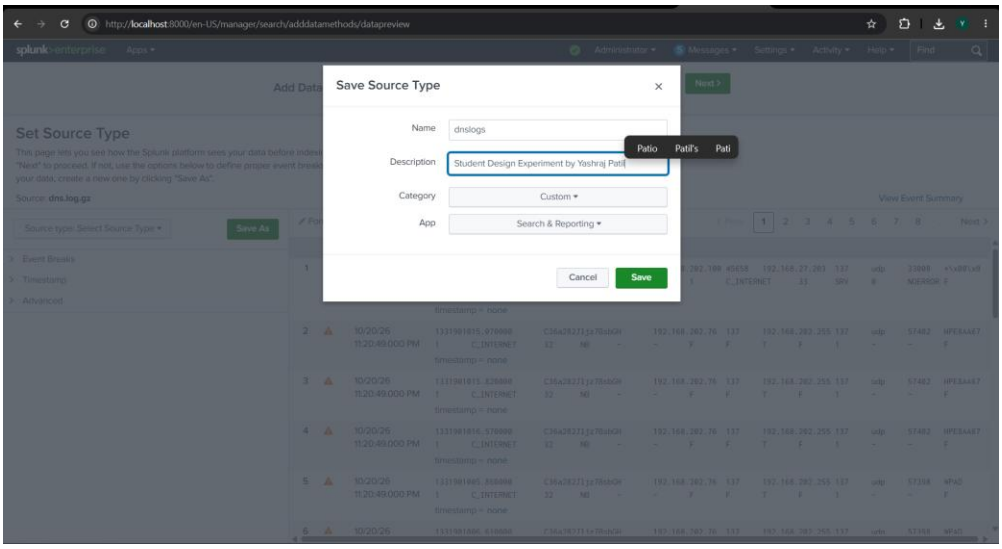

splunk>enterprise
Apps
Administrator
Messages
Settings
Activity
Help
Find
What data do you want to send to the Splunk platform?
Follow guides for onboarding popular data sources
Search bar
Cloud computing
Get your cloud computing data in to the Splunk platform.
10 data sources
Networking
Get your networking data in to the Splunk platform.
2 data sources
Operating System
Get your operating system data in to the Splunk platform.
1 data source
Security
Get your security data in to the Splunk platform.
3 data sources
4 data sources in total
Or get data in with the following methods
Upload
files from my computer
Monitor
files and ports on this Splunk platform instance
Forward
data from a Splunk forwarder

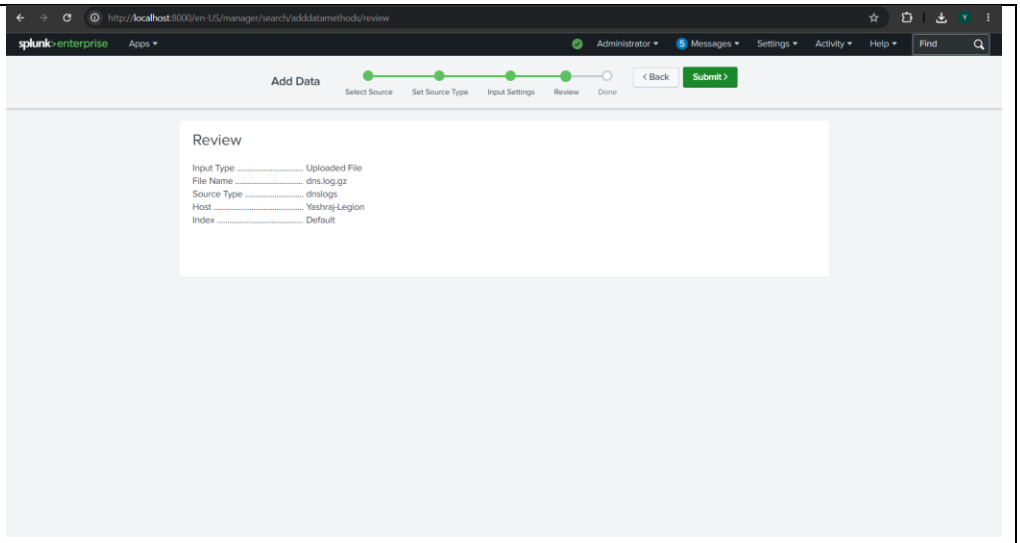

splunk>enterprise
Apps
Administrator
Messages
Settings
Activity
Help
Find
Add Data
Select Source
Set Source Type
Input Settings
Review
Done
< Back
Next >
Select Source
Choose a file to upload to the Splunk platform, either by browsing your computer or by dropping a file into the target box below. Learn More
Selected File: No file selected
Select File
Drop your data file here
The maximum file upload size is 500 Mb
FAQ
What kinds of files can the Splunk platform index?
What is a source?
How do I get remote data onto my Splunk platform instance?

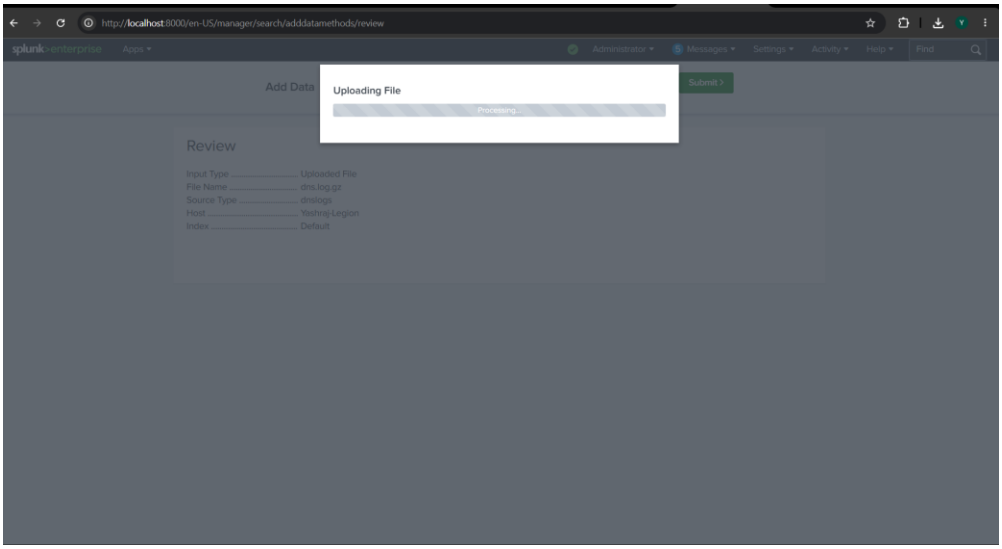

esults in Downloads
dns.log.gz
Name
Date modified
Type
Size
Earlier this year
dns.log.gz
18-06-2026 22:23
WinRAR
6,453 KB
Type: WinRAR
Size: 6.30 MB
Date modified: 18-06-2026 22:23
Path: dns.log (C:\Users\Yashraj\Downloads)
Search again in
Libraries
All Files
Open
Cancel

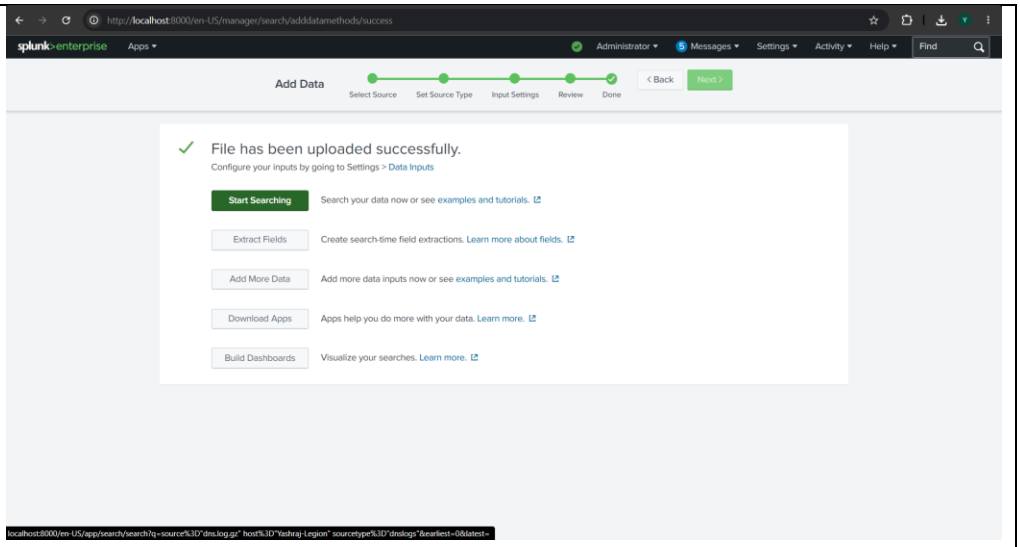

splunk>enterprise
Apps
Administrator
Messages
Settings
Activity
Help
Find
Add Data
Select Source
Set Source Type
Input Settings
Review
Done
Back
Next
Select Source
Choose a file to upload to the Splunk platform, either by browsing your computer or by dropping a file into the target box below. Learn More
Selected File: dns.log.gz
Select File
Drop your data file here
The maximum file upload size is 500 Mb
File Successfully Uploaded
FAQ
What kinds of files can the Splunk platform index?
What is a source?
How do I get remote data onto my Splunk platform instance?

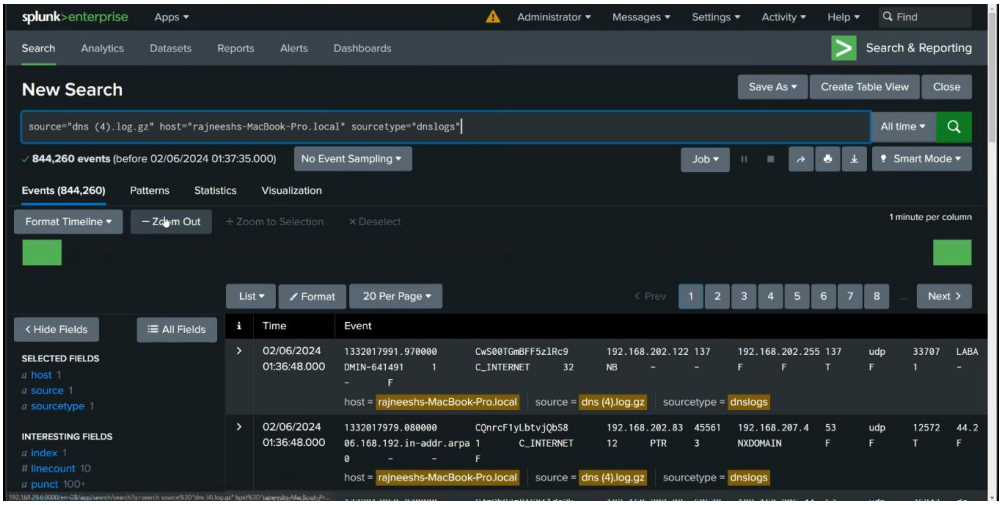

Add Data
Select Source
Set Source Type
Input Settings
Review
Done
Back
Next
Set Source Type
This page lets you see how the Splunk platform sees your data before indexing. If the events look correct and have the right timestamps, click "Next" to proceed. If not, use the options below to define proper event breaks and timestamps. If you cannot find an appropriate source type for your data, create a new one by clicking "Save As".
Source: dns.log.gz
View Event Summary
Source type: Select Source Type
Save As
Event Breaks
Timestamp
Advanced
Time
Event

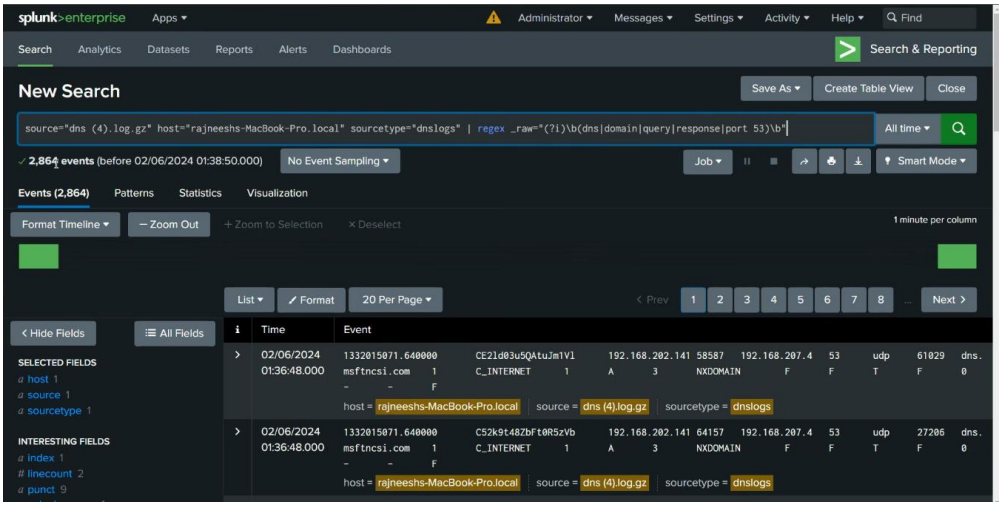

Save Source Type
Name
dnslogs
Description
Student Design Experiment by Yashraj Patil
Category
Custom
App
Search & Reporting
Cancel
Save
Set Source Type

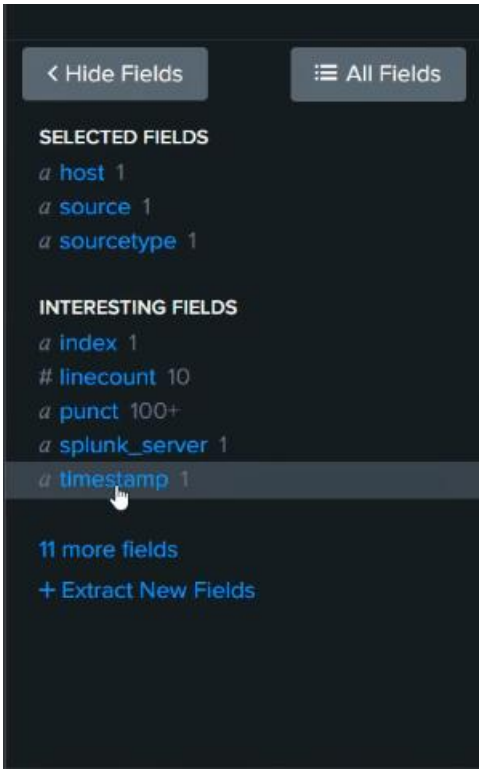

splunk>enterprise
Apps
Administrator
Messages
Settings
Activity
Help
Find
Add Data
Select Source
Set Source Type
Input Settings
Review
Done
Back
Submit
Review
Input Type ..... Uploaded File
File Name ..... dns.log.gz
Source Type ..... dnslogs
Host ..... Yashraj-Legion
Index ..... Default

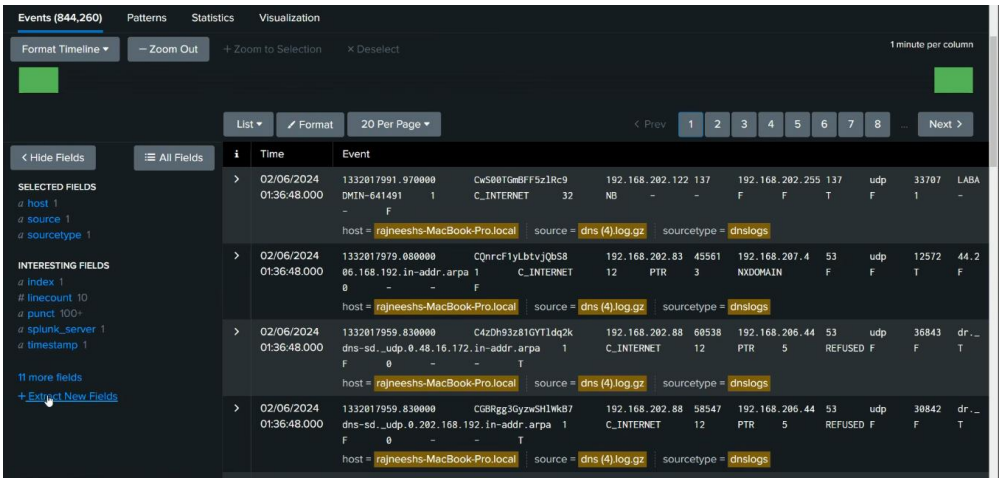

Uploading File
Processing...
Review

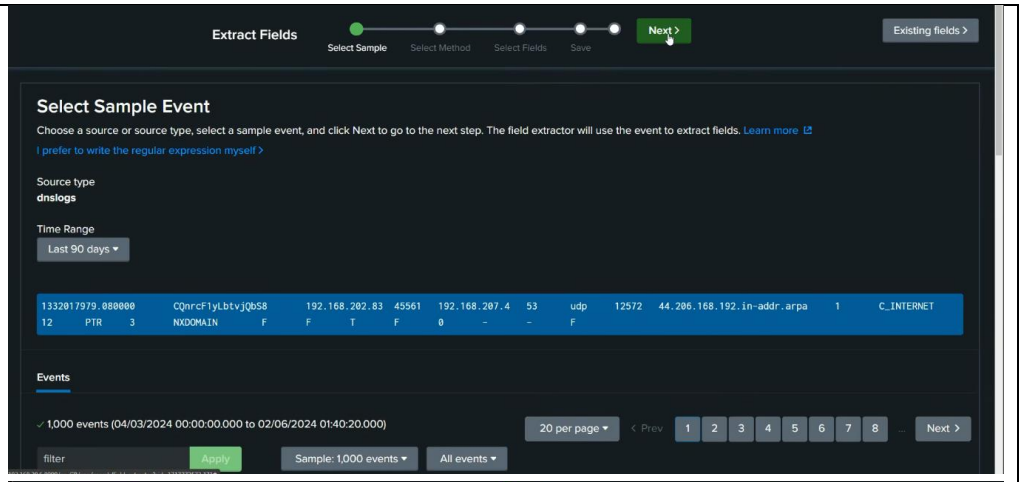

splunk>enterprise
Add Data
File has been uploaded successfully.
Configure your inputs by going to Settings > Data Inputs
Start Searching
Search your data now or see examples and tutorials.
Extract Fields
Create search-time field extractions. Learn more about fields.
Add More Data
Add more data inputs now or see examples and tutorials.
Download Apps
Apps help you do more with your data. Learn more.
Build Dashboards
Visualize your searches. Learn more.

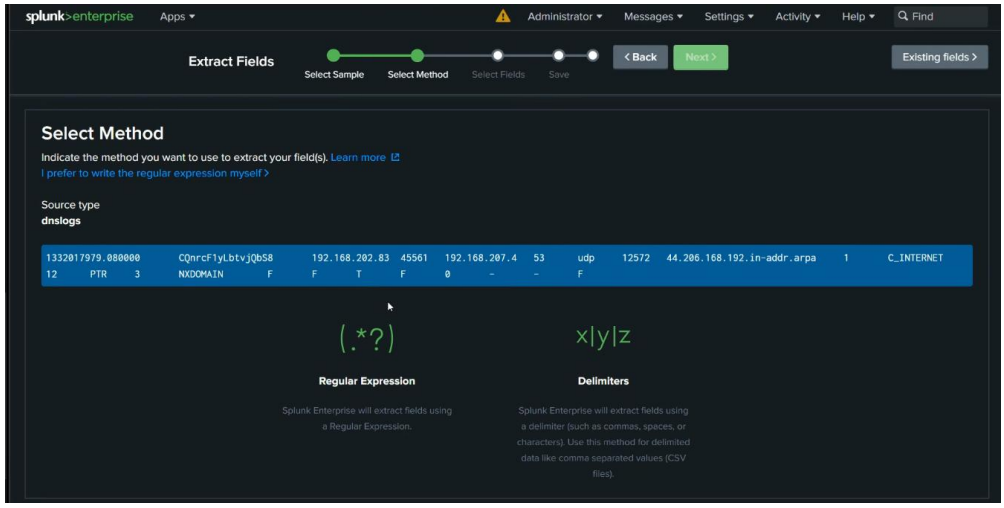

splunk>enterprise
New Search
source="dns (4).log.gz" host="rajneeshs-MacBook-Pro.local" sourcetype="dnslogs"
844,260 events (before 02/06/2024 01:37:35.000)
Events (844,260)
Patterns
Statistics
Visualization
Time
Event
host = rajneeshs-MacBook-Pro.local
source = dns (4).log.gz
sourcetype = dnslogs

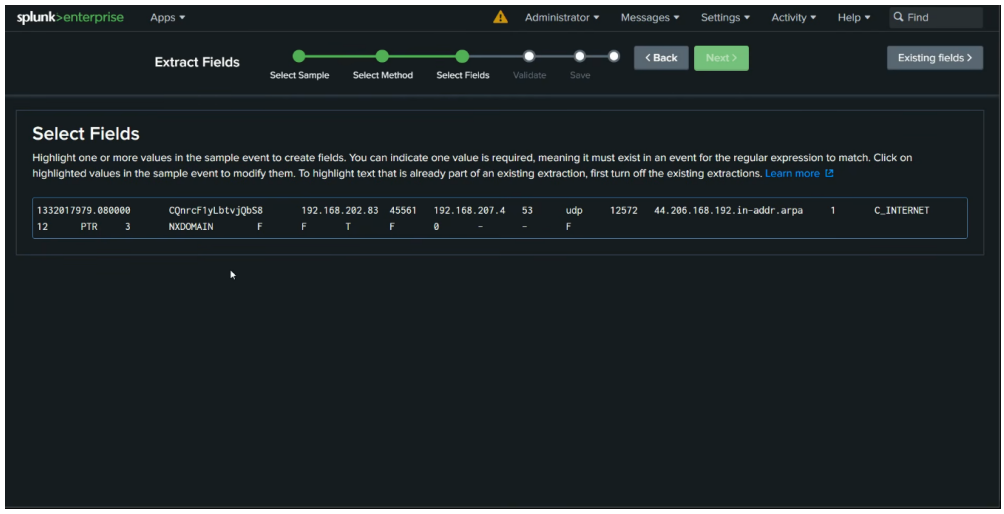

splunk>enterprise
New Search
source="dns (4).log.gz" host="rajneeshs-MacBook-Pro.local" sourcetype="dnslogs" | regex _raw="(?i)\b(dns|domain|query|response|port 53)\b"
2,864 events (before 02/06/2024 01:38:50.000)
Events (2,864)
Patterns
Statistics
Visualization
Time
Event
host = rajneeshs-MacBook-Pro.local
source = dns (4).log.gz
sourcetype = dnslogs

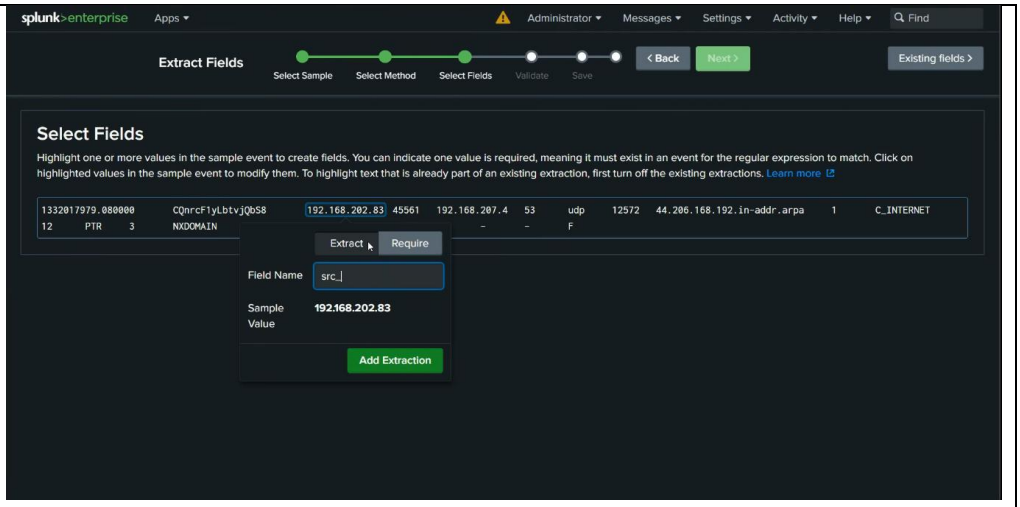

Hide Fields
All Fields
SELECTED FIELDS
a host 1
a source 1
a sourcetype 1
INTERESTING FIELDS
a index 1
# linecount 10
a punct 100+
a splunk_server 1
a timestamp 1
11 more fields
+ Extract New Fields

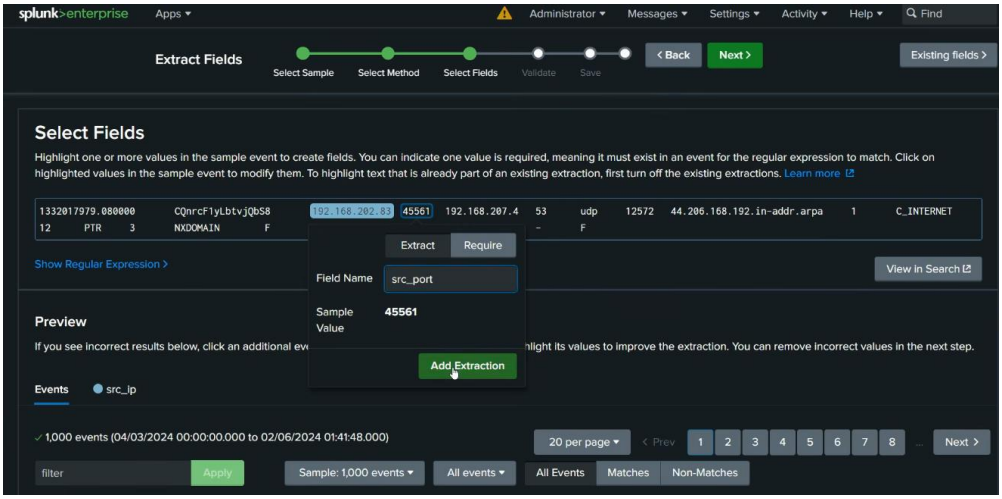

Events (844,260)
Patterns
Statistics
Visualization
Format Timeline
Zoom Out
Zoom to Selection
Deselect
1 minute per column
List
Format
20 Per Page
Prev
Next
Hide Fields
All Fields
Time
Event
SELECTED FIELDS
a host 1
a source 1
a sourcetype 1
INTERESTING FIELDS
a index 1
# linecount 10
a punct 100+
a splunk_server 1
a timestamp 1
11 more fields
+ Extract New Fields
02/06/2024 01:36:48.000
host = rajneeshs-MacBook-Pro.local
source = dns (4).log.gz
sourcetype = dnslogs

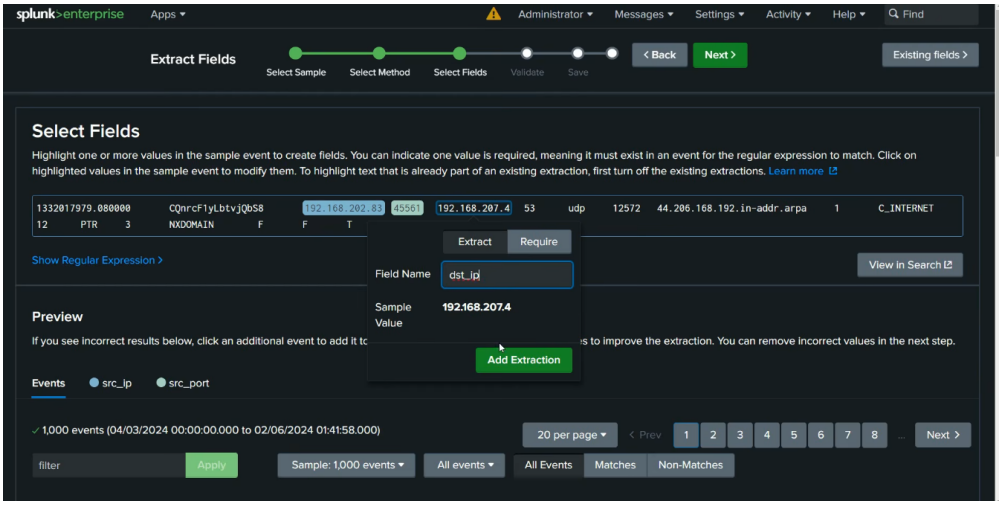

Extract Fields
Select Sample
Select Method
Select Fields
Save
Next >
Existing fields >
Select Sample Event
Choose a source or source type, select a sample event, and click Next to go to the next step. The field extractor will use the event to extract fields. Learn more
I prefer to write the regular expression myself >
Source type
dnslogs
Time Range
Last 90 days
1332017979.080000 CQnrcF1yLbtvjQbS8 192.168.202.83 45561 192.168.207.4 53 udp 12572 44.206.168.192.in-addr.arpa 1 C_INTERNET
12 PTR 3 NXDOMAIN F F T F 0 - - F
Events
1,000 events (04/03/2024 00:00:00.000 to 02/06/2024 01:40:20.000)
20 per page
Prev
1 2 3 4 5 6 7 8
Next >
filter
Apply
Sample: 1,000 events
All events

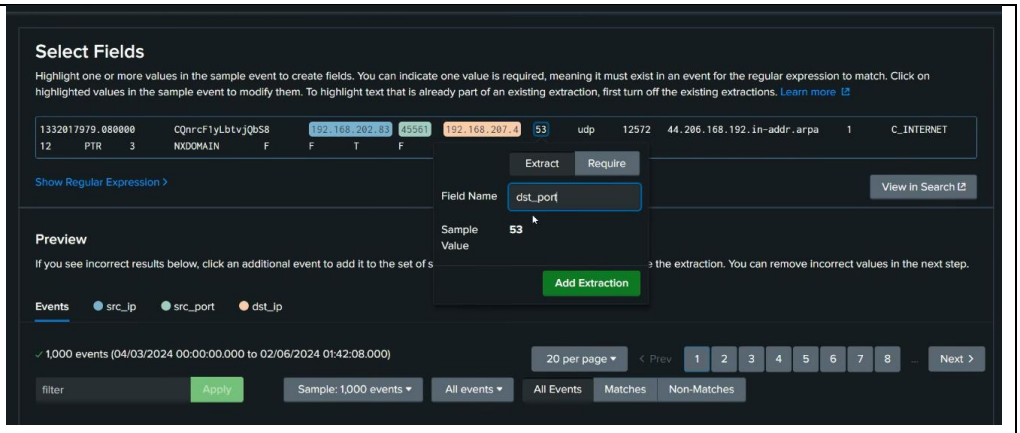

splunk>enterprise
Apps
Administrator
Messages
Settings
Activity
Help
Find
Extract Fields
Select Sample
Select Method
Select Fields
Save
< Back
Next >
Existing fields >
Select Method
Indicate the method you want to use to extract your field(s). Learn more
I prefer to write the regular expression myself >
Source type
dnslogs
1332017979.080000 CQnrcF1yLbtvjQbS8 192.168.202.83 45561 192.168.207.4 53 udp 12572 44.206.168.192.in-addr.arpa 1 C_INTERNET
12 PTR 3 NXDOMAIN F F T F 0 - - F
(.*?)
Regular Expression
Splunk Enterprise will extract fields using a Regular Expression.
x|y|z
Delimiters
Splunk Enterprise will extract fields using a delimiter (such as commas, spaces, or characters). Use this method for delimited data like comma separated values (CSV files).

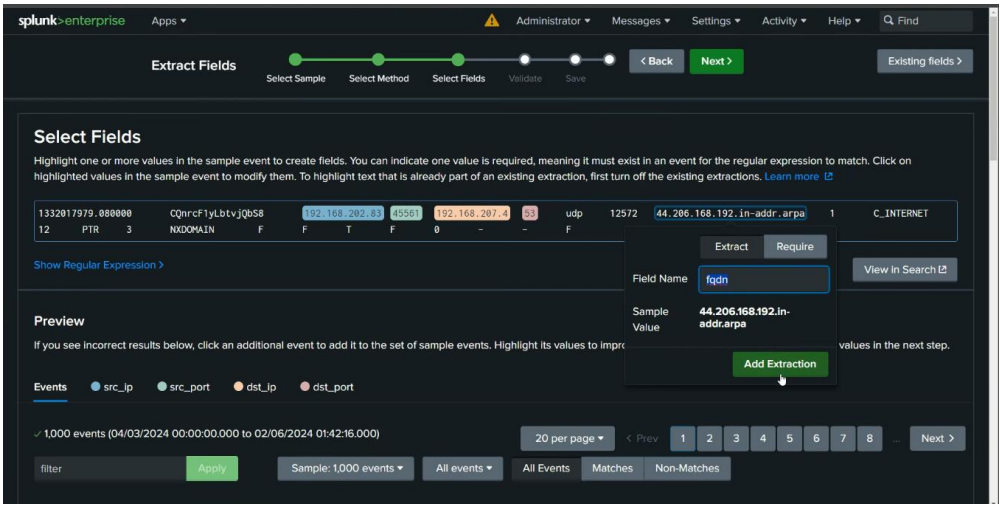

splunk>enterprise
Apps
Administrator
Messages
Settings
Activity
Help
Find
Extract Fields
Select Sample
Select Method
Select Fields
Validate
Save
< Back
Next >
Existing fields >
Select Fields
Highlight one or more values in the sample event to create fields. You can indicate one value is required, meaning it must exist in an event for the regular expression to match. Click on highlighted values in the sample event to modify them. To highlight text that is already part of an existing extraction, first turn off the existing extractions. Learn more
1332017979.080000 CQnrcF1yLbtvjQbS8 192.168.202.83 45561 192.168.207.4 53 udp 12572 44.206.168.192.in-addr.arpa 1 C_INTERNET
12 PTR 3 NXDOMAIN F F T F 0 - - F

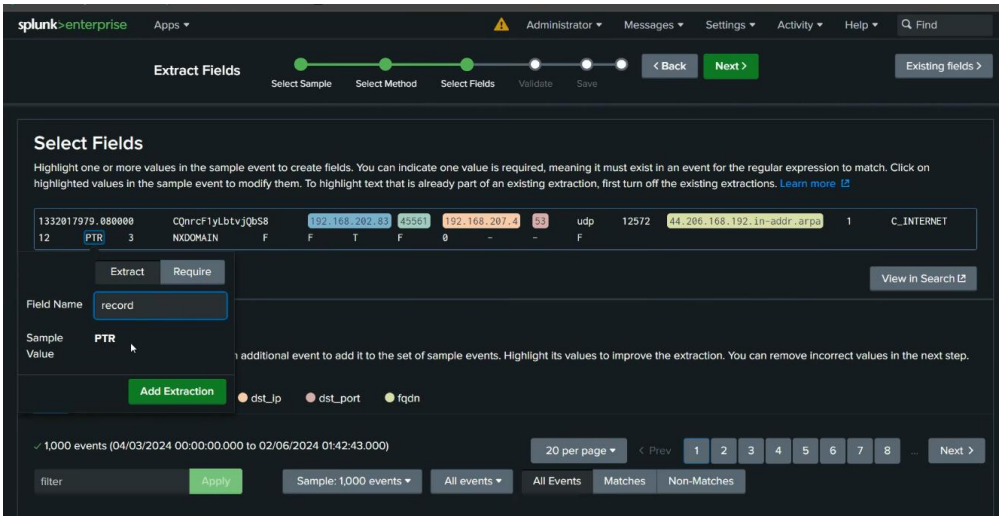

splunk>enterprise
Apps
Administrator
Messages
Settings
Activity
Help
Find
Extract Fields
Select Sample
Select Method
Select Fields
Validate
Save
Back
Next
Existing fields
Select Fields
Highlight one or more values in the sample event to create fields. You can indicate one value is required, meaning it must exist in an event for the regular expression to match. Click on highlighted values in the sample event to modify them. To highlight text that is already part of an existing extraction, first turn off the existing extractions. Learn more
1332017979.080000 CQnrcF1yLbtvjQbS8 192.168.202.83 45561 192.168.207.4 53 udp 12572 44.206.168.192.in-addr.arpa 1 C_INTERNET
12 PTR 3 NXDOMAIN
Extract
Require
Field Name
src_
Sample Value
192.168.202.83
Add Extraction

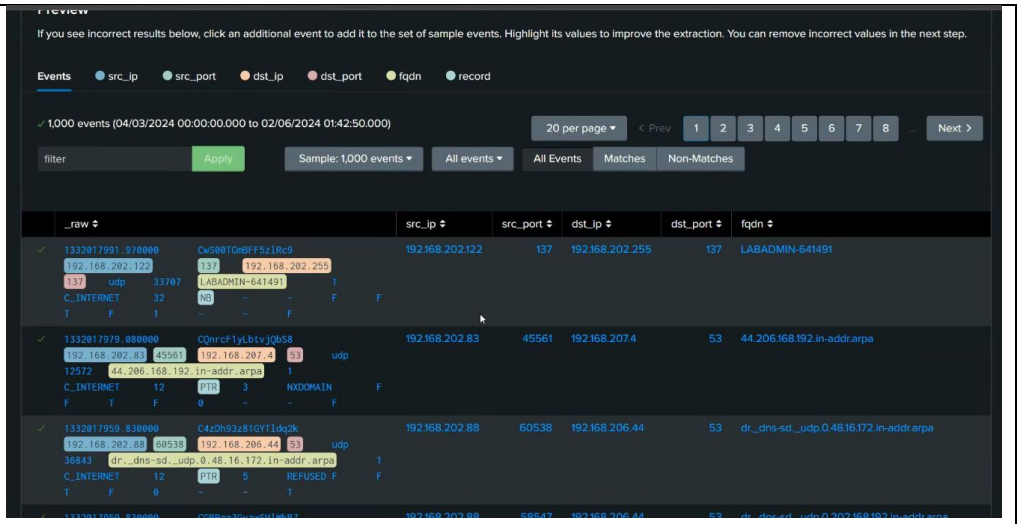

splunk>enterprise
Apps
Administrator
Messages
Settings
Activity
Help
Find
Extract Fields
Select Sample
Select Method
Select Fields
Validate
Save
Back
Next
Existing fields
Select Fields
Highlight one or more values in the sample event to create fields. You can indicate one value is required, meaning it must exist in an event for the regular expression to match. Click on highlighted values in the sample event to modify them. To highlight text that is already part of an existing extraction, first turn off the existing extractions. Learn more
1332017979.080000 CQnrcF1yLbtvjQbS8 192.168.202.83 45561 192.168.207.4 53 udp 12572 44.206.168.192.in-addr.arpa 1 C_INTERNET
12 PTR 3 NXDOMAIN F
Extract
Require
Show Regular Expression
View in Search
Field Name
src_port
Sample Value
45561
Add Extraction
Preview
If you see incorrect results below, click an additional ev
hlight its values to improve the extraction. You can remove incorrect values in the next step.
Events
src_ip
1,000 events (04/03/2024 00:00:00.000 to 02/06/2024 01:41:48.000)
20 per page
Prev
1 2 3 4 5 6 7 8
Next
filter
Apply
Sample: 1,000 events
All events
All Events
Matches
Non-Matches

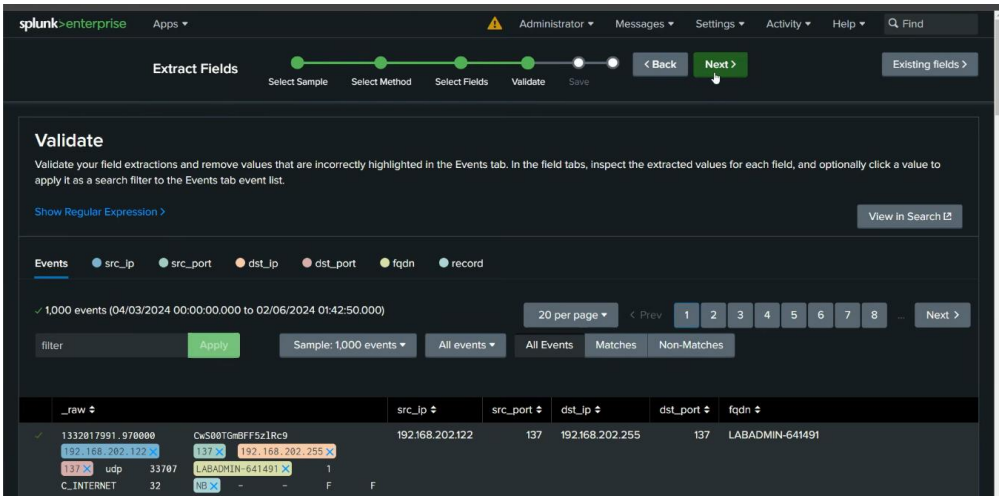

splunk>enterprise
Apps
Administrator
Messages
Settings
Activity
Help
Find
Extract Fields
Select Sample
Select Method
Select Fields
Validate
Save
Back
Next
Existing fields
Select Fields
Highlight one or more values in the sample event to create fields. You can indicate one value is required, meaning it must exist in an event for the regular expression to match. Click on highlighted values in the sample event to modify them. To highlight text that is already part of an existing extraction, first turn off the existing extractions. Learn more
1332017979.080000 CQnrcF1yLbtvjQbS8 192.168.202.83 45561 192.168.207.4 53 udp 12572 44.206.168.192.in-addr.arpa 1 C_INTERNET
12 PTR 3 NXDOMAIN F F T
Extract
Require
Show Regular Expression
View in Search
Field Name
dst_ip
Sample Value
192.168.207.4
Add Extraction
Preview
If you see incorrect results below, click an additional event to add it tc
s to improve the extraction. You can remove incorrect values in the next step.
Events
src_ip
src_port
1,000 events (04/03/2024 00:00:00.000 to 02/06/2024 01:41:58.000)
20 per page
Prev
1 2 3 4 5 6 7 8
Next
filter
Apply
Sample: 1,000 events
All events
All Events
Matches
Non-Matches

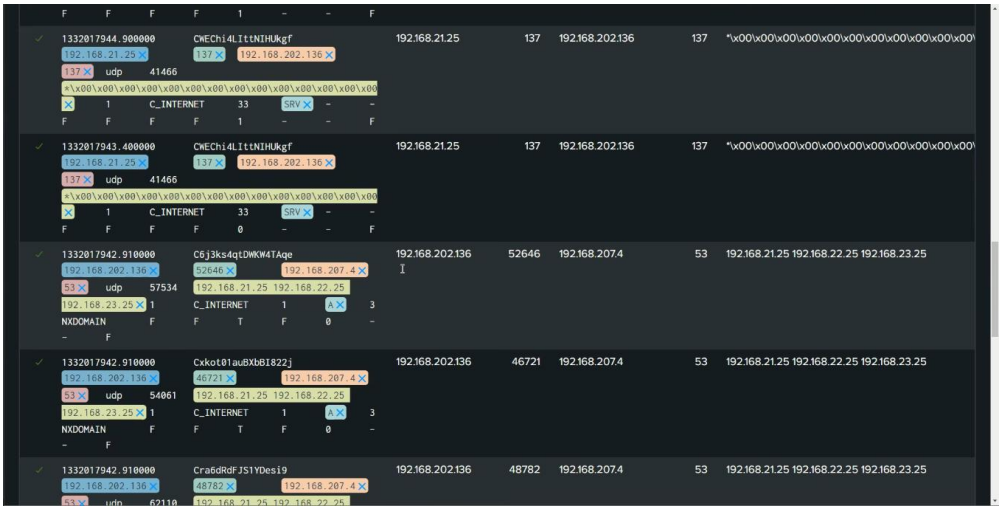

Select Fields
Highlight one or more values in the sample event to create fields. You can indicate one value is required, meaning it must exist in an event for the regular expression to match. Click on highlighted values in the sample event to modify them. To highlight text that is already part of an existing extraction, first turn off the existing extractions. Learn more
1332017979.080000 CQnrcF1yLbtvjQbS8 192.168.202.83 45561 192.168.207.4 53 udp 12572 44.206.168.192.in-addr.arpa 1 C_INTERNET
12 PTR 3 NXDOMAIN F F T F
Extract Require
Field Name dst_port
Sample Value 53
Add Extraction
Show Regular Expression
View in Search
Preview
If you see incorrect results below, click an additional event to add it to the set of s the extraction. You can remove incorrect values in the next step.
Events src_ip src_port dst_ip
1,000 events (04/03/2024 00:00:00.000 to 02/06/2024 01:42:08.000)
20 per page Prev 1 2 3 4 5 6 7 8 Next
filter Apply Sample: 1,000 events All events All Events Matches Non-Matches

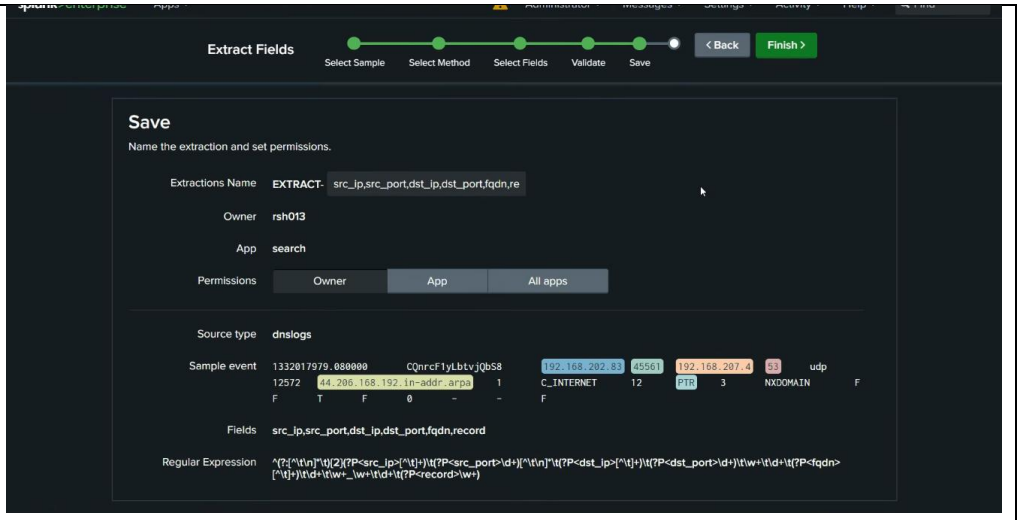

splunk>enterprise Apps Administrator Messages Settings Activity Help Find
Extract Fields
Select Sample Select Method Select Fields Validate Save
Back Next Existing fields
Select Fields
Highlight one or more values in the sample event to create fields. You can indicate one value is required, meaning it must exist in an event for the regular expression to match. Click on highlighted values in the sample event to modify them. To highlight text that is already part of an existing extraction, first turn off the existing extractions. Learn more
1332017979.080000 CQnrcF1yLbtvjQbS8 192.168.202.83 45561 192.168.207.4 53 udp 12572 44.206.168.192.in-addr.arpa 1 C_INTERNET
12 PTR 3 NXDOMAIN F F T F 0 - - F
Extract Require
Field Name fqdn
Sample Value 44.206.168.192.in-addr.arpa
Add Extraction
Show Regular Expression
View in Search
Preview
If you see incorrect results below, click an additional event to add it to the set of sample events. Highlight its values to impr values in the next step.
Events src_ip src_port dst_ip dst_port
1,000 events (04/03/2024 00:00:00.000 to 02/06/2024 01:42:16.000)
20 per page Prev 1 2 3 4 5 6 7 8 Next
filter Apply Sample: 1,000 events All events All Events Matches Non-Matches

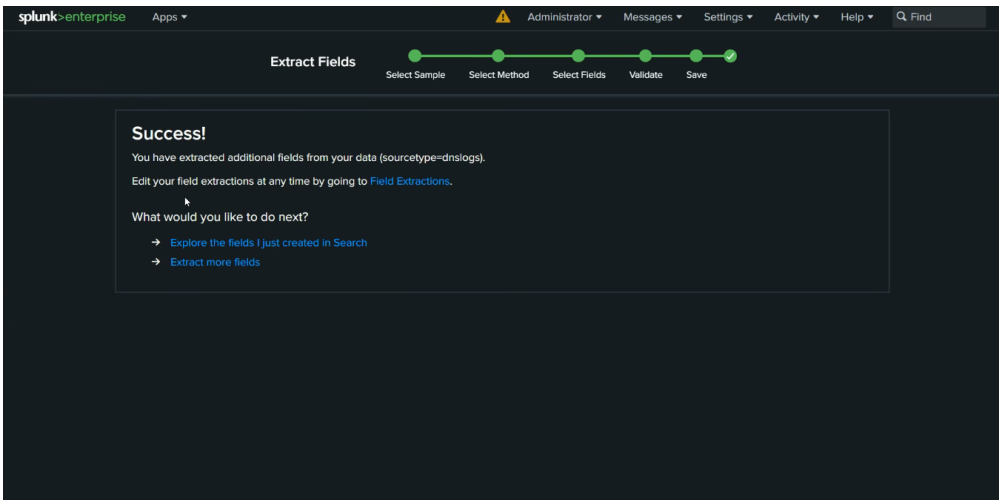

splunk>enterprise Apps Administrator Messages Settings Activity Help Find
Extract Fields
Select Sample Select Method Select Fields Validate Save
Back Next Existing fields
Select Fields
Highlight one or more values in the sample event to create fields. You can indicate one value is required, meaning it must exist in an event for the regular expression to match. Click on highlighted values in the sample event to modify them. To highlight text that is already part of an existing extraction, first turn off the existing extractions. Learn more
1332017979.080000 CQnrcF1yLbtvjQbS8 192.168.202.83 45561 192.168.207.4 53 udp 12572 44.206.168.192.in-addr.arpa 1 C_INTERNET
12 PTR 3 NXDOMAIN F F T F 0 - - F
Extract Require
Field Name record
Sample Value PTR
Add Extraction
View in Search
additional event to add it to the set of sample events. Highlight its values to improve the extraction. You can remove incorrect values in the next step.
dst_ip dst_port fqdn
1,000 events (04/03/2024 00:00:00.000 to 02/06/2024 01:42:43.000)
20 per page Prev 1 2 3 4 5 6 7 8 Next
filter Apply Sample: 1,000 events All events All Events Matches Non-Matches

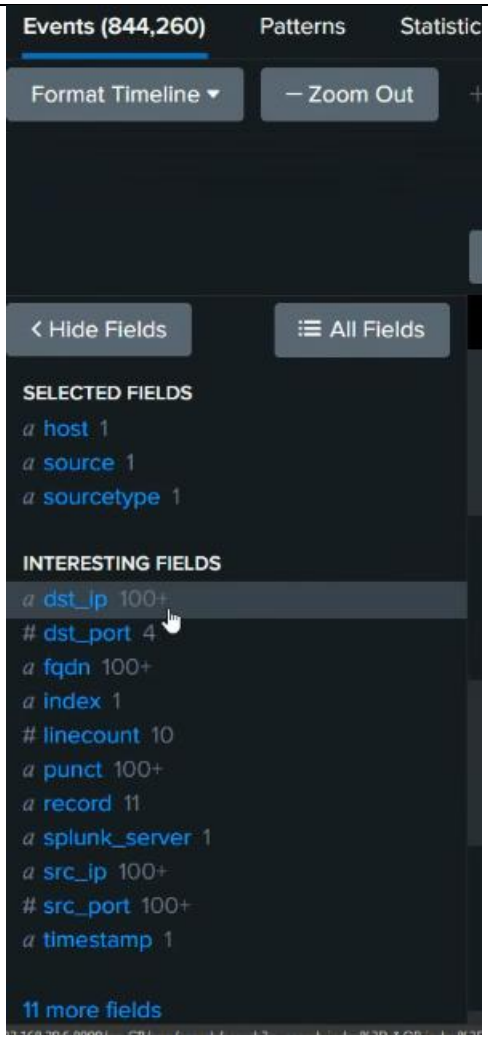

If you see incorrect results below, click an additional event to add it to the set of sample events. Highlight its values to improve the extraction. You can remove incorrect values in the next step.
Events
src_ip
src_port
dst_ip
dst_port
fqdn
record
1,000 events (04/03/2024 00:00:00.000 to 02/06/2024 01:42:50.000)
20 per page
Prev
Next
filter
Apply
Sample: 1,000 events
All events
All Events
Matches
Non-Matches
_raw
src_ip
src_port
dst_ip
dst_port
fqdn
1332017991.970000 CwS00TGmBFF5zlRc9 192.168.202.122 137 192.168.202.255 137 LABADMIN-641491
1332017979.080000 CQnrcF1yLbtvjQbS8 192.168.202.83 45561 192.168.207.4 53 44.206.168.192.in-addr.arpa
1332017959.830000 C4zDh93z81GYTldq2k 192.168.202.88 60538 192.168.206.44 53 dr._dns-sd._udp.0.48.16.172.in-addr.arpa

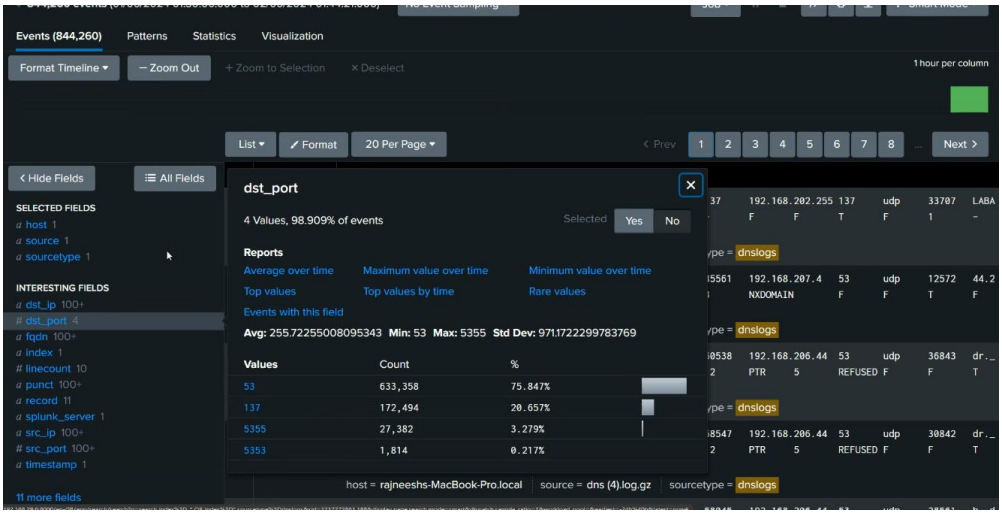

splunk>enterprise
Apps
Administrator
Messages
Settings
Activity
Help
Find
Extract Fields
Select Sample
Select Method
Select Fields
Validate
Save
< Back
Next >
Existing fields >
Validate
Validate your field extractions and remove values that are incorrectly highlighted in the Events tab. In the field tabs, inspect the extracted values for each field, and optionally click a value to apply it as a search filter to the Events tab event list.
Show Regular Expression >
View in Search
Events
src_ip
src_port
dst_ip
dst_port
fqdn
record
1,000 events (04/03/2024 00:00:00.000 to 02/06/2024 01:42:50.000)
20 per page
Prev
Next
filter
Apply
Sample: 1,000 events
All events
All Events
Matches
Non-Matches
_raw
src_ip
src_port
dst_ip
dst_port
fqdn
1332017991.970000 CwS00TGmBFF5zlRc9 192.168.202.122 137 192.168.202.255 137 LABADMIN-641491

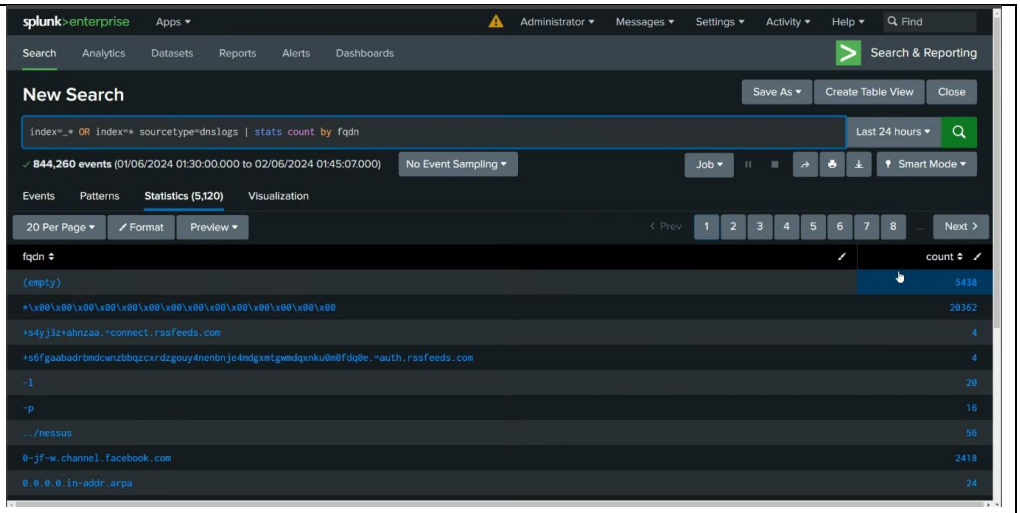

1332017944.900000 CWEChi4LIttNIHUkgf 192.168.21.25 137 192.168.202.136 137
1332017943.400000 CWEChi4LIttNIHUkgf 192.168.21.25 137 192.168.202.136 137
1332017942.910000 C6j3ksdqtDWKW4TAqe 192.168.202.136 52646 192.168.207.4 53 192.168.21.25 192.168.22.25 192.168.23.25
1332017942.910000 Cxkot01auBXbBI822j 192.168.202.136 46721 192.168.207.4 53 192.168.21.25 192.168.22.25 192.168.23.25
1332017942.910000 Cra6dRdFJS1YDesi9 192.168.202.136 48782 192.168.207.4 53 192.168.21.25 192.168.22.25 192.168.23.25

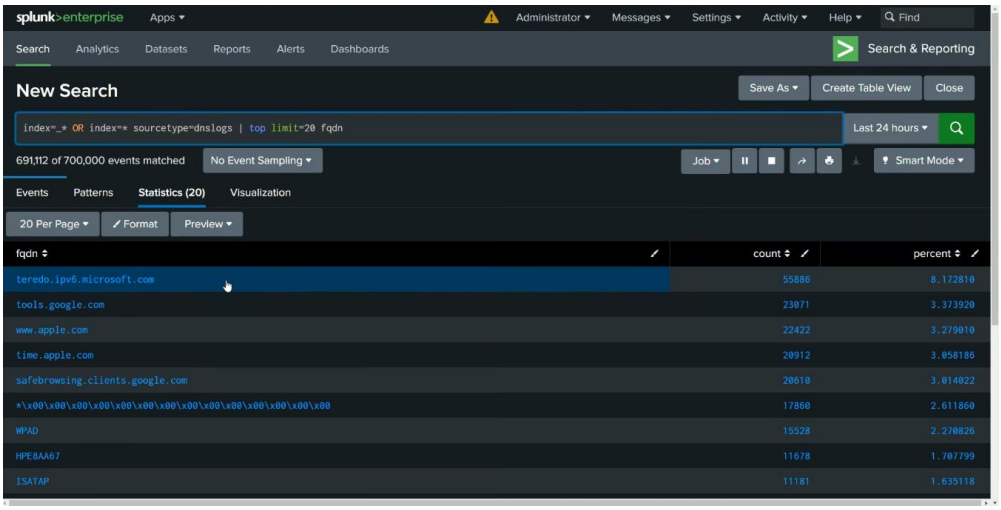

Extract Fields
Select Sample
Select Method
Select Fields
Validate
Save
< Back
Finish >
Save
Name the extraction and set permissions.
Extractions Name
EXTRACT- src_ip,src_port,dst_ip,dst_port,fqdn,re
Owner
rsh013
App
search
Permissions
Owner
App
All apps
Source type
dnslogs
Sample event
Fields
src_ip,src_port,dst_ip,dst_port,fqdn,record
Regular Expression

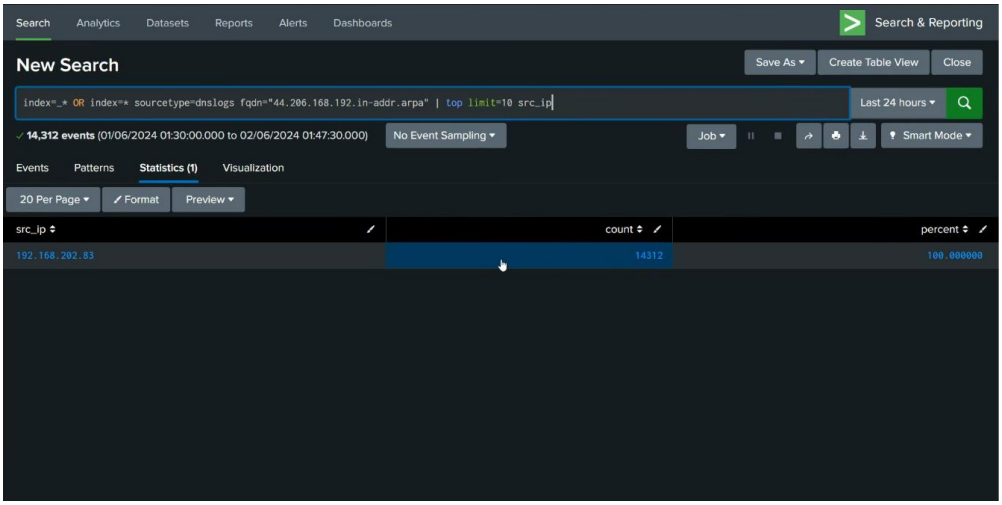

splunk>enterprise
Apps
Administrator
Messages
Settings
Activity
Help
Find
Extract Fields
Select Sample
Select Method
Select Fields
Validate
Save
Success!
You have extracted additional fields from your data (sourcetype=dnslogs).
Edit your field extractions at any time by going to Field Extractions.
What would you like to do next?
Explore the fields I just created in Search
Extract more fields

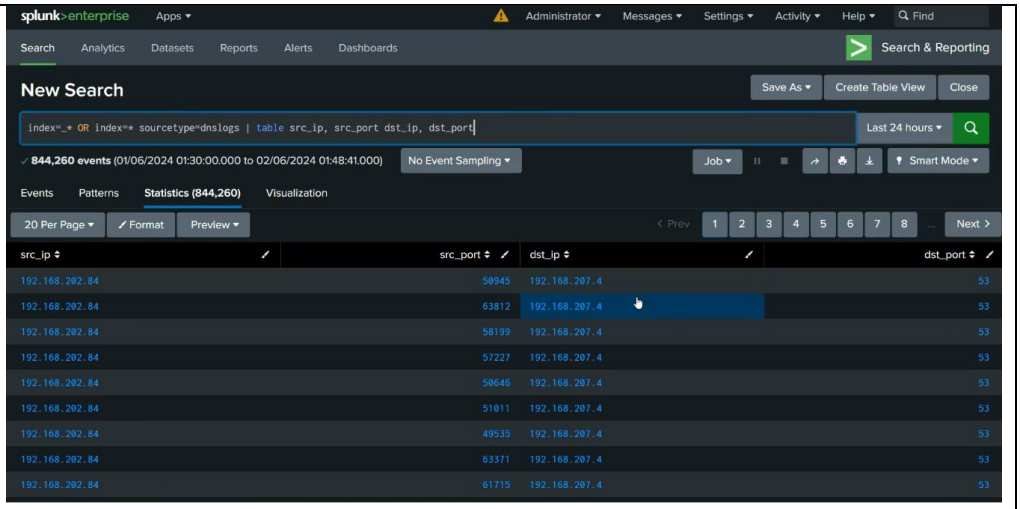

Events (844,260)
Patterns
Statistic
Format Timeline
– Zoom Out
Hide Fields
All Fields
SELECTED FIELDS
a host 1
a source 1
a sourcetype 1
INTERESTING FIELDS
a dst_ip 100+
# dst_port 4
a fqdn 100+
a index 1
# linecount 10
a punct 100+
a record 11
a splunk_server 1
a src_ip 100+
# src_port 100+
a timestamp 1
11 more fields


Events (844,260)
Patterns
Statistics
Visualization
Format Timeline
– Zoom Out
+ Zoom to Selection
× Deselect
1 hour per column
List
Format
20 Per Page
Prev
Next
Hide Fields
All Fields
SELECTED FIELDS
INTERESTING FIELDS
dst_port
4 Values, 98.909% of events
Selected
Yes
No
Reports
Average over time
Maximum value over time
Minimum value over time
Top values
Top values by time
Rare values
Events with this field
Avg: 255.72255008095343 Min: 53 Max: 5355 Std Dev: 971.1722299783769
Values
Count
%
53
633,358
75.847%
137
172,494
20.657%
5355
27,382
3.279%
5353
1,814
0.217%
host = rajneeshs-MacBook-Pro.local
source = dns (4).log.gz
sourcetype = dnslogs
11 more fields


splunk>enterprise
Apps
Administrator
Messages
Settings
Activity
Help
Find
Search
Analytics
Datasets
Reports
Alerts
Dashboards
Search & Reporting
New Search
Save As
Create Table View
Close
index=_* OR index=* sourcetype=dnslogs | stats count by fqdn
Last 24 hours
844,260 events (01/06/2024 01:30:00.000 to 02/06/2024 01:45:07.000)
No Event Sampling
Job
Smart Mode
Events
Patterns
Statistics (5,120)
Visualization
20 Per Page
Format
Preview
Prev
1
2
3
4
5
6
7
8
Next
fqdn
count
(empty) 5438
*\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00 20362
+s4yj3z+ahnzaa.~connect.rssfeeds.com 4
+s6fgaabadrbmdcwnzbbqzcxrdzgouy4nenbnje4mdgxmtgwmdqxnku0m0fdq0e.~auth.rssfeeds.com 4
-l 20
-p 16
../nessus 56
0-jf-w.channel.facebook.com 2418
0.0.0.0.in-addr.arpa 24


splunk>enterprise
Apps
Administrator
Messages
Settings
Activity
Help
Find
Search
Analytics
Datasets
Reports
Alerts
Dashboards
Search & Reporting
New Search
Save As
Create Table View
Close
index=_* OR index=* sourcetype=dnslogs | top limit=20 fqdn
Last 24 hours
691,112 of 700,000 events matched
No Event Sampling
Job
Smart Mode
Events
Patterns
Statistics (20)
Visualization
20 Per Page
Format
Preview
fqdn
count
percent
teredo.ipv6.microsoft.com 55886 8.172810
tools.google.com 23071 3.373920
www.apple.com 22422 3.279010
time.apple.com 20912 3.058186
safebrowsing.clients.google.com 20610 3.014022
*\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00 17860 2.611860
WPAD 15528 2.270826
HPE8AA67 11678 1.707799
ISATAP 11181 1.635118


Search
Analytics
Datasets
Reports
Alerts
Dashboards
Search & Reporting
New Search
Save As
Create Table View
Close
index=_* OR index=* sourcetype=dnslogs fqdn="44.206.168.192.in-addr.arpa" | top limit=10 src_ip
Last 24 hours
14,312 events (01/06/2024 01:30:00.000 to 02/06/2024 01:47:30.000)
No Event Sampling
Job
Smart Mode
Events
Patterns
Statistics (1)
Visualization
20 Per Page
Format
Preview
src_ip
count
percent
192.168.202.83 14312 100.000000




**Conclusion**

This lab showed how DNS logs can be used to detect web application reconnaissance using Splunk SIEM. Sample DNS logs were uploaded and indexed under the dns_sample source type, and their key fields (source IP, destination IP, domain name, query type and response code) were extracted and searched.

SPL commands such as regex, stats, top and timechart made it easy to count domain queries, find the most active hosts and spot sudden bursts of activity. These helped identify signs of reconnaissance, including subdomain scanning, many failed lookups and requests to suspicious domains. Unfamiliar domains were then investigated and checked against threat intelligence sources like VirusTotal.

Overall, the experiment shows that DNS monitoring gives early warning of attacker activity, and that a SIEM like Splunk turns raw logs into useful security insight. These skills form a practical foundation for threat hunting and work in a Security Operations Center (SOC).